Mixed boundary data for the Poisson equation

First-order versus ghost-point closures of a five-point scheme

TEAM F

Kunze Li, 2363973 — Project Manager
Yawen Wu, 2360749 — Mathematical Theory
Che Li, 2362072 — Algorithm Implementation
Zekun Wang, 2363718 — Visualization & Report
Yixuan Liu, 2361140 — Testing & Validation
Ruitong Huang, 2360294 — Video & presentation

Xi'an Jiaotong-Liverpool University

The problem: Dirichlet data on three edges, a flux law on the fourth

On $\Omega = (0,1)^2$ let $\Gamma_B = \{1\} \times (0,1)$ be the open right edge and $\Gamma_D = \partial\Omega \setminus \Gamma_B$ (owns all four corners). Seek u with

$$-\Delta u = f \text{ in } \Omega, \qquad u = g \text{ on } \Gamma_D, \qquad \partial_n u + \beta u = r \text{ on } \Gamma_B, \qquad \partial_n u = u_x.$$

Neumann: $\beta = 0$, $r = q$. Robin: $\beta = 2$.

Manufactured solution (Problem Pack): $u^* = e^{x+y}$, $f = -2e^{x+y}$, $g = u^*|_{\Gamma_D}$, $q = e^{1+y}$, $r = (1+\beta)e^{1+y}$. Sign check $q > 0$: outward normal, u^* increases towards Γ_B.

Question (Direction 2). How does the flux-edge closure decide the global order of a five-point scheme, and what does it do to the linear system?

Steady heat conduction with convective Robin exchange (Bergman–Incropera et al., 2011); pressure-Poisson with Neumann wall data (Gresho–Sani, 1987).

Well-posedness: Lax–Milgram, provided $\beta \geq 0$

With $V = \{v \in H^1(\Omega) : v|_{\Gamma_D} = 0\}$ and a lifting g_* of g: find $u \in g_* + V$ with

$$a(u,v) = \int_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x + \beta \int_{\Gamma_B} uv \,\mathrm{d}s = \int_\Omega fv \,\mathrm{d}x + \int_{\Gamma_B} rv \,\mathrm{d}s = \ell(v) \qquad \forall v \in V.$$

Proposition. *If $f \in L^2(\Omega)$, $r \in L^2(\Gamma_B)$, $g = \operatorname{tr} g_*$ for $g_* \in H^1(\Omega)$ and $\beta \geq 0$ constant, the weak problem has a unique solution depending continuously on (f, r, g_*).*

- Boundedness: Cauchy–Schwarz and the trace inequality $\|v\|_{L^2(\Gamma_B)} \leq C_T \|v\|_{H^1}$.
- Coercivity: Γ_D has positive length ⇒ Friedrichs on V with $C_P = 1/\sqrt{2}$; for $\beta \geq 0$, $a(v,v) \geq \|\nabla v\|^2 \geq (1 + C_P^2)^{-1} \|v\|_{H^1}^2$.
- Sign: coercivity uses $\beta \int_{\Gamma_B} v^2 \geq 0$, so only $\beta \geq 0$ is claimed. Γ_D has positive measure ⇒ no compatibility condition.

Orders below assume $u \in C^4$ up to Γ_B; mixed problems may lose regularity at $(1,0)$, $(1,1)$ (Grisvard). Not claimed for nonsmooth data.

Discretisation: one stencil, two closures of the flux edge

$h = 1/N$; unknowns $U_{i,j}$, $1 \le i \le N$, $1 \le j \le N-1$ ($M = N(N-1)$; Dirichlet values to the RHS). Interior, $1 \le i \le N-1$:

$$(4U_{i,j} - U_{i-1,j} - U_{i+1,j} - U_{i,j-1} - U_{i,j+1})/h^2 = f_{i,j}.$$

Closure 1, first-order row. Replace the PDE row at (N, j) by a backward difference of the BC:

$$\left(\tfrac{1}{h} + \beta\right)U_{N,j} - \tfrac{1}{h}U_{N-1,j} = r_j.$$

Closure 2, ghost point. Impose $(U_{N+1,j} - U_{N-1,j})/(2h) + \beta U_{N,j} = r_j$ and eliminate $U_{N+1,j}$:

$$\left(\tfrac{4}{h^2} + \tfrac{2\beta}{h}\right)U_{N,j} - \tfrac{2}{h^2}U_{N-1,j} - \tfrac{1}{h^2}(U_{N,j-1} + U_{N,j+1}) = f_{N,j} + \tfrac{2}{h}r_j.$$

Closure 1 enforces only the BC (scale h^{-1}); closure 2 keeps the PDE with the BC built in (scale h^{-2}).

Residuals compared only in the row-scaled form $\rho = \max_k |(F - AU)_k|/(|A||U| + |F|)_k$, invariant under these scalings.

Truncation error: both boundary rows are only $O(h)$

With $\tau = A\,u^*|_{\text{grid}} - F$ and Taylor at $(1, y_j)$:

$$\tau_{i,j} = -\frac{h^2}{12}\,(u_{xxxx} + u_{yyyy}) + o(h^2) \quad (u \in C^4),$$

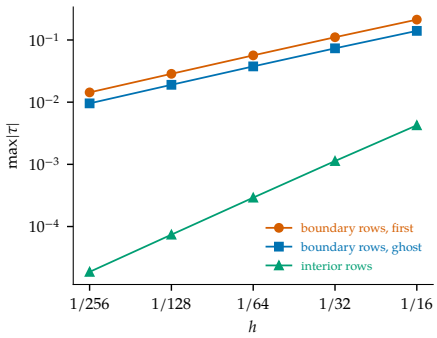
$$\tau^{(1)}_{N,j} = -\frac{h}{2}\,u_{xx}(1, y_j) + O(h^2),$$

$$\tau^{(2)}_{N,j} = \frac{h}{3}\,u_{xxx}(1, y_j) + O(h^2).$$

For u^*: $\max|\tau_B|/h \to e^2/2$ (first), $e^2/3$ (ghost); $\max|\tau_I|/h^2 \to e^2/6$. Local order alone does not explain different global orders.

Robin, $N = 256$	$\max\|\tau_B\|/h$	predicted	$\max\|\tau_I\|/h^2$
first-order	3.6753	3.6945	1.2219
ghost-point	2.4486	2.4630	1.2219

Substituting u^* into the assembled rows.



Robin, $\max|\tau|$ vs h: both boundary rows $\sim h$, interior $\sim h^2$.

Matrix structure: DA symmetric positive definite

Proposition. *Let $D = \operatorname{diag}(d_k)$ with $d_k = 1$ on five-point rows and, on right-edge rows, $d_k = 1/h$ (first-order) or $d_k = \frac{1}{2}$ (ghost). Then DA is symmetric, and positive definite for $\beta \geq 0$.*

The only nonsymmetric pairs couple $(N-1, j)$ and (N, j): $-h^{-2}$ against $-h^{-1}$ (first) or $-2h^{-2}$ (ghost); D makes both $-h^{-2}$. For the ghost closure

$$h^2 v^{\mathrm{T}} DAv = \sum_{\text{horiz.}} (v_p - v_q)^2 + \sum_{\text{vert.},\, x<1} (v_p - v_q)^2 + \tfrac{1}{2} \sum_{\text{vert.},\, x=1} (v_p - v_q)^2 + \beta h \sum_j v_{N,j}^2,$$

the half-cell weights are trapezoidal: DA is the finite-volume form of the weak problem.

A itself is nonsymmetric: CG applies to $DAU = DF$, not to A merely because $-\Delta$ is self-adjoint.

Matrix structure: nonsingular M-matrix

THEOREM (*M-matrix*). *For $\beta \geq 0$ and either closure, A is a nonsingular M-matrix, so $A^{-1} \geq 0$ (discrete maximum principle).*

Positive diagonal, nonpositive off-diagonals; every row weakly diagonally dominant. Rows coupling to eliminated Dirichlet nodes are strict; right-edge rows additionally strict when $\beta > 0$. First-order rows at $\beta = 0$ are only weakly dominant, yet A remains irreducibly diagonally dominant (Varga).

Robin	$\eta(A)$, $N = 16/64/256$	$\eta(DA)$	ρ
first	0.0769 / 0.0393 / 0.0197	0	$\leq 5.4 \times 10^{-13}$
ghost	0.0789 / 0.0395 / 0.0198	0	$\leq 7.2 \times 10^{-16}$

$\eta(X) = \|X - X^{\mathrm{T}}\|_F / \|X\|_F$.

$\eta(DA) = 0$ on these meshes: diagonal scaling restores symmetry to machine precision.

Max-norm error bounds via comparison functions

$e = U - u^*|_{\text{grid}}$ solves $Ae = -\tau$, $\tau = \tau_I + \tau_B$. Comparison functions $\phi = \frac{1}{2}x(2-x) \le \frac{1}{2}$, $\psi = x \le 1$. Baseline row values (exact for full stencils; reduced A gains $C_v \ge 0$ near $y = 0, 1$):

	interior rows	first-order rows	ghost rows
$A\phi \ \ge$	1	$\frac{1}{2}(h+\beta) \ge 0$	$1 + \beta/h \ge 1$
$A\psi \ \ge$	0	$1 + \beta \ge 1$	$2(1+\beta)/h \ge 2/h$

Theorem. *Let $\beta \ge 0$ and $u \in C^4(\bar{\Omega})$. Then*

$$\textit{first-order: } \|e\|_\infty \le \tfrac{1}{2}\|\tau_I\|_\infty + \|\tau_B^{(1)}\|_\infty = O(h), \qquad \textit{ghost: } \|e\|_\infty \le \tfrac{1}{2}\|\tau_I\|_\infty + \tfrac{h}{2}\|\tau_B^{(2)}\|_\infty = O(h^2).$$

Mechanism ($A\psi$): both $\tau_B = O(h)$, but $A\psi \ge 1$ on first-order rows vs. $A\psi \ge 2/h$ on ghost rows (h^{-2} scaling). Covering $|\tau_B|$ needs $\|\tau_B\|_\infty$ vs. $\frac{h}{2}\|\tau_B\|_\infty$ of $\psi \le 1$: $O(h)$ vs. $O(h^2)$. With $w = \|\tau_I\|_\infty \phi + c\|\tau_B\|_\infty \psi$ ($c = 1$ or $h/2$), $Aw \ge |\tau| \Rightarrow |e| \le w$ by the M-matrix lemma.

Our comparison-function derivation. Background: LeVeque Ch. 2–3; Morton–Mayers (2005) Ch. 6.

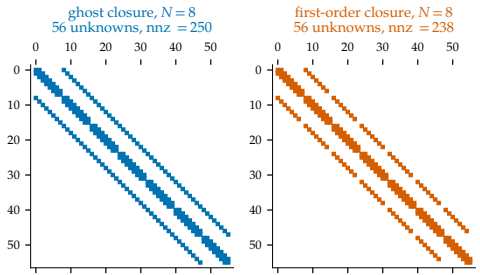
Code validation: manufactured solution, five mesh levels

Protocol. $u^\star = e^{x+y}$; $N = 16, 32, 64, 128, 256$; $p = \log_2(E_h/E_{h/2})$. Norms: nodal max, and $\|e\|_{L_h^2}^2 = h^2 \sum_k w_k e_k^2$ with $w_k = \frac{1}{2}$ on right-edge nodes.

Assembly. COO→CSR, index $k = (j-1)N + (i-1)$; scipy.sparse.linalg.spsolve.

Checks. (i) τ matches slide 5. (ii) $\rho \le 8.1 \times 10^{-13}$. (iii) CG on $DAU = DF$, $N = 64$: 309–324 it.; $\|U_{\mathrm{CG}} - U_{\mathrm{direct}}\| \le 1.2 \times 10^{-11}$. (iv) $\kappa_2(DA) \approx 2.18 \times 10^3$ (Robin, ghost, $N = 64$).

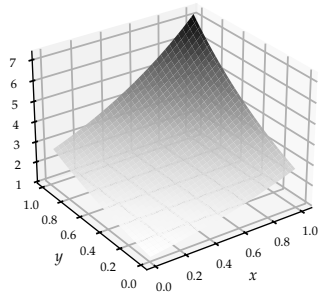
Bank P1–P5. Ghost order 2.00 (P2: error at round-off, $\approx 7 \times 10^{-13}$, no meaningful order); first 1.03–1.11, except P1/P4 2.04–2.05 ($u_{xx}(1, y) = 0$).



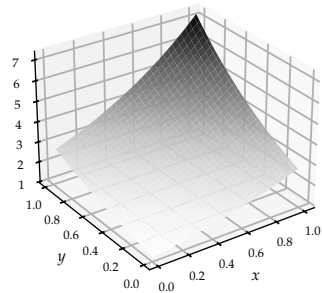
Sparsity of A, $N = 8$: ghost 250 nnz vs first-order 238; differ only on right-edge rows.

Solution surface: exact, numerical, error

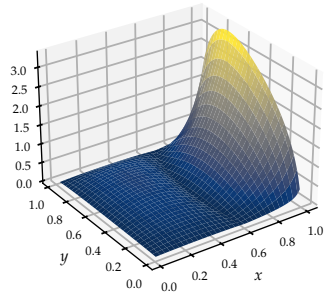
exact $u^* = e^{x+y}$



numerical U, $N = 64$



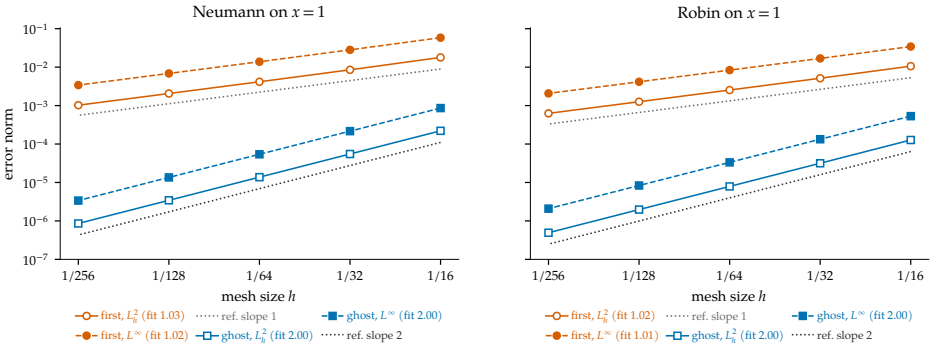
error $|U - u^*|$ ($\times 10^{-5}$)



Robin ($\beta = 2$), ghost closure, $N = 64$. $|U - u^*| \leq 3.34 \times 10^{-5}$ on Γ_B; vanishes at Dirichlet edges.

U matches $u^* = e^{x+y}$; the flux-edge closure sets the accuracy.

Convergence: orders ≈ 1 and 2, in both norms



Fit, all levels: L^2_h order 1.03/2.00 (Neu.), 1.02/2.00 (Rob.); max log-dev ≤ 0.016.

Asymptotic from $N = 32$: pairwise $p \in [1.00, 1.07]$ and $[1.99, 2.01]$. Max error on Γ_B.

Observed orders and errors versus bounds

	closure	p, L_h^2	p, L^∞
Neumann	first	1.01	1.01
	ghost	2.00	2.00
Robin	first	1.00	1.00
	ghost	2.00	2.00

Last pair $N = 128 \to 256$.

$N = 256$	$\|e\|_\infty$ Neumann	bound	$\|e\|_{\infty,\mathrm{int}}$ Neumann	$\|e\|_\infty$ Robin
first-order	3.411×10^{-3}	$3.69h \approx 1.4 \times 10^{-2}$	3.373×10^{-3}	2.075×10^{-3}
ghost-point	3.385×10^{-6}	$1.85h^2 \approx 2.8 \times 10^{-5}$	3.334×10^{-6}	2.085×10^{-6}

- Observed max errors sit below the comparison-function bounds, with factors ≈ 4 (first-order) and ≈ 8 (ghost).
- Interior max error is almost as large as the global max: the flux-edge closure limits first-order accuracy.

Bounds from the $A\psi$ comparison on slide 8; orders proved $O(h)$ / $O(h^2)$, observed 1.00–1.01 / 2.00.

Cost versus accuracy: matched target $\|e\|_\infty \le 5 \times 10^{-3}$

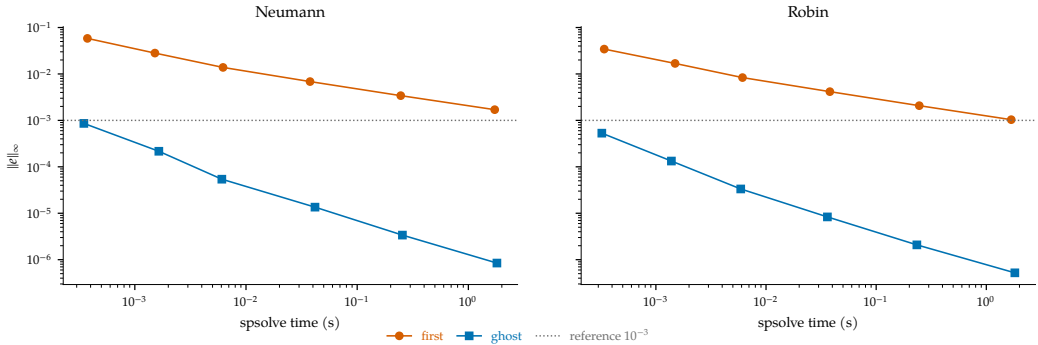
Cost: median spsolve time, five runs (assembly, I/O, CG, plots excluded).

BC	Closure	$N_{\min}$ (tested)	unknowns	$\|e\|_\infty$	spsolve (s)
Neumann	first-order	256	65280	3.411×10^{-3}	2.46×10^{-1}
Neumann	ghost-point	16	240	8.611×10^{-4}	3.48×10^{-4}
Robin	first-order	128	16256	4.159×10^{-3}	3.80×10^{-2}
Robin	ghost-point	16	240	5.307×10^{-4}	3.23×10^{-4}

- Ghost meets the target at $N = 16$ (coarsest tested); first-order needs $N = 128$ (Robin) or $N = 256$ (Neumann). Solve-time ratio $\sim 10^2$–10^3.
- Dotted line on the next slide: tighter reference 10^{-3} (not the table target); first-order Neumann at $N = 512$ still 1.70×10^{-3}.

$N_{\min}$: smallest tested mesh meeting the target, $N \in \{16, 32, 64, 128, 256, 512\}$. Source: results/cost_accuracy.csv.

Cost curves: $\|e\|_\infty$ versus spsolve time



Dotted line: reference 10^{-3} (stricter than the table target 5×10^{-3}). Ghost reaches the table target at $N = 16$.

Same timings as the cost table; assembly, I/O, CG and plots excluded.

Limitations

- One smooth manufactured solution on the unit square. Orders need C^4 up to Γ_B; not claimed for nonsmooth or incompatible corner data, or for $\beta < 0$.
- Algebraic vs discretisation error: $\rho \leq 8.1 \times 10^{-13}$; CG/direct $\leq 1.2 \times 10^{-11}$ at $N = 64$; $\kappa_2(DA) \approx 2.18 \times 10^3$ (Robin, ghost, $N = 64$), growing like $O(N^2)$.
- Two check families: MMS substitution, and course analytic bank P1–P5.
- Cost–accuracy at $\|e\|_\infty \leq 5 \times 10^{-3}$ (spsolve only); 10^{-3} not reached by first-order within $N \leq 512$.
- Package under code/ is vectorised.
- For this manufactured solution, Robin errors are smaller than Neumann at the same mesh resolution.

Optional depth (pure-Neumann nullspace and gauge) not attempted.

Conclusion and reproducibility

- Comparison functions prove $\|e\|_\infty = O(h)$ (first) and $O(h^2)$ (ghost); observed fitted orders ($N = 16$–256) are 1.01–1.03 and 2.00 in both norms.
- The flux-edge closure, not the interior stencil, limits first-order accuracy.
- Both A are nonsymmetric; diagonal scaling makes DA SPD and legitimises CG.
- At $\|e\|_\infty \leq 5 \times 10^{-3}$ the ghost closure is 10^2–10^3 times cheaper in spsolve time.

Reproduce. From project2/code/:

command	produces
python run_all.py	convergence, MMS checks, cost, figures, tables
python convergence.py	results/convergence.csv
python mms_runner.py	truncation, κ, CG cross-check
python cost_accuracy.py	matched-target timings

Code and data: github.com/AXE111111111/321F/tree/main/Project 2.